

Alberto Reyes Castro

Applied AI & ML Engineer | Multimodal, On-Device & LLM Systems

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Portfolio | GitHub | LinkedIn | Kaggle

UK work authorisation: Student visa valid until May 2028; up to 20 hours/week during term time and full-time during official university vacations.

Professional Summary

Applied AI engineer and Physics graduate completing an MSc in Artificial Intelligence Technology with Advanced Practice, with current academic performance at Distinction level. Building and evaluating machine-learning systems across vision-language retrieval, on-device inference, LLM-assisted developer tools, speech processing and scientific computing. Experienced in taking projects from research and benchmarking through implementation, validation, technical documentation and human-reviewed product workflows.

Technical Skills

AI, machine learning and multimodal systems: Python, PyTorch, TensorFlow/Keras (coursework), Hugging Face Transformers, scikit-learn, XGBoost, vision-language models, image-text retrieval, embeddings, fine-tuning, imbalanced learning, cross-validation, threshold analysis and Recall@K.

On-device, mobile and software engineering: ONNX Runtime, React Native, Expo, Kotlin, SQLite/SQL, TypeScript, Next.js, React, API integration, Git/GitHub, Docker (foundational), Bash/PowerShell, Linux/Ubuntu, Jupyter/Colab and debugging.

LLMs, speech and reliable AI workflows: Gemini API, Llama 3.2, Ollama, local LLM workflows, deterministic and provider fallbacks, faster-whisper, multilingual transcription, SRT generation, regex and fuzzy matching.

Data, cloud and technical communication: pandas, NumPy, SciPy, Matplotlib, R/RStudio (coursework), Google Cloud Platform (GCP) and cloud-architecture fundamentals, LaTeX/Overleaf, Markdown, technical reports, scientific writing and reproducible notebooks.

AIoT and embedded prototyping: C++/Arduino, ESP32-S3, Wokwi, Blynk, Edge Impulse/TinyML and sensor-based automation.

Relevant Experience

Applied AI Engineer - On-Device Multimodal ML

TeachScribe - Technical Collaboration

Jul 2026 - Present

London, UK (Hybrid)

- Built and evaluated a standalone Android prototype for potential adaptation into TeachScribe's teacher-facing EdTech product, ranking classroom photographs against written observations through optional capture-time filtering and fully on-device image-text inference.
- Designed three structured retrieval evaluations and benchmarked five pretrained vision-language models using R@1 and R@5 alongside model size, CPU performance, mobile feasibility and published checkpoint licensing; identified TinyCLIP as the provisional candidate.
- Integrated TinyCLIP through ONNX Runtime into an Expo/React Native prototype with native Kotlin preprocessing, reusable sessions, cosine-similarity ranking and persistent SQLite embedding caches; implemented progress/cancellation and human-reviewed multi-photo selection.
- Current work includes task-specific fine-tuning and validation on the target Android tablet, evaluating retrieval quality, latency, memory use and thermal behaviour against the pretrained baseline.

Creator & Applied AI Developer - ModelHawk

Independent AI/ML Developer-Tool Prototype

Jul 2026 - Present

London, UK (Remote)

- Designed and built a Next.js/TypeScript prototype currently focused on structured classification notebooks, particularly imbalanced datasets; parses notebook code and Markdown and applies deterministic and heuristic checks for leakage risk, metric choice, validation quality and reproducibility.
- Integrated optional examiner notes through the Gemini API with a locally hosted Llama 3.2 model through Ollama as fallback, while retaining deterministic operation without an LLM; generates structured findings, reviewer questions and professional LaTeX PDF reports.

Python & Scientific Communication Intern

Instituto de Astrofísica de Canarias (IAC)

Mar 2025 - May 2025

La Laguna, Spain (On-site)

- Developed a portable Python workflow for multilingual audio transcription using faster-whisper, automatic CPU/GPU selection and segmented SRT generation, reducing manual transcription and subtitle-preparation work.
- Built a human-verified matching pipeline comparing 67 outreach articles with more than 700 scientific publications using preprocessing, regular expressions and fuzzy matching.
- Authored the IAC Vía Láctea article *Eclipse Solar Parcial de Marzo 2025*, wrote the English-language version of an IAC article on SN 2023ixf and co-authored an *Astronomía* Magazine feature; completed the assessed 250-hour placement with 9.2/10.

Selected Projects

Credit Card Fraud Detection - Imbalanced Machine Learning

- Built a leakage-controlled pipeline for 284,807 transactions, comparing six classifiers and multiple imbalance strategies using five-fold stratified cross-validation, RandomizedSearchCV and validation-based threshold analysis.
- Tuned XGBoost achieved 91.2% precision, 84.7% recall and 0.8827 Average Precision. MSc project mark: 85/100.

AIoT Smart Home Prototype - Embedded and TinyML

- Built two ESP32-S3 simulations in Arduino/C++ using Wokwi, Blynk and Edge Impulse, integrating sensor-driven control, actuators, safety-state logic and lightweight environmental classification.

Computational Physics Numerical Methods

- Implemented interpolation, numerical integration, ODE solvers and stochastic simulation in Python using NumPy, SciPy and Matplotlib, with reproducible notebook demonstrations.

Open-Source Contribution

Vizro/Dash Rendering Regression Analysis

- Reproduced and isolated a tab-rendering regression by comparing dependency versions, testing minimal pure-Dash behaviour and separating the issue from Plotly graph rendering; the analysis was acknowledged by a Vizro maintainer and helped narrow the source of the regression.

Selected Publications & Scientific Communication

- *Eclipse Solar Parcial de Marzo 2025* - authored for the IAC Vía Láctea science-outreach blog during the internship.
- *The WEAVE spectrograph captures the interaction of supernova SN 2023ixf with its surroundings one year after its explosion* - wrote the English-language version for IAC News.
- *El eclipse parcial que recordó a Canarias el valor de mirar al cielo* - co-authored with Ariadna Mesa and Hugo Fernández; *Astronomía Magazine*, No. 311 (May 2025).

Education

MSc Artificial Intelligence Technology with Advanced Practice **Jan 2026 - Jan 2028**
Northumbria University, London Campus *London, UK*

Current academic performance: Distinction level (70%+). First-semester results of 70, 75 and 85, including 85/100 for an end-to-end machine-learning project on credit-card fraud detection. Relevant modules include Programming for AI, Machine Learning on Cloud, Artificial Intelligence for IoT, Information Governance and Cyber Security, and Professional Practices in Computing.

BSc in Physics **Sep 2018 - Jul 2025**
Universidad de La Laguna *Tenerife, Spain*

Final grade: 7.13/10 (approximately equivalent to a UK 2:1). Academic Merit Convocation Award recipient. Selected results: General Relativity (10/10); Acquisition and Transmission of Signals and Images (9.9); Differential Calculus (9.5); Electronic Physics (9.0); Statistical Physics (8.7); Nuclear and Particle Physics (8.2); Differential Equations and Complex Variables (8.1); Remote Sensing (8.1). Final Degree Project completed and defended in English with 9.4/10; the supervisor subsequently proposed developing it into a manuscript for submission to a high-impact peer-reviewed scientific journal.

Languages & Selected Training

English: C1 - IELTS Academic UKVI 7.0/9.0 and Cambridge Advanced Certificate. **Spanish:** Native. **Japanese:** Intermediate study. **French:** Limited working proficiency.

AI & Big Data for the Biomedical and Healthcare Sector (300 hours), DYM Canarias, 2025: Python data analysis, machine learning, TensorFlow/Keras, Docker and APIs, Big Data/cloud architecture, and probability/statistics with R.

Google Cloud Skills Boost, 2025: Innovating with Google Cloud Artificial Intelligence; Exploring Data Transformation with Google Cloud; Digital Transformation with Google Cloud.

Additional relevant training: NASA Open Science 101 (2025); Introduction to Data Science, IE University/Santander Open Academy (2024); Artificial Intelligence and Productivity, Google/Santander Open Academy (2024).